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PHARMACOLOGY

Individual Components of Polymyxin B Modeled via Population Pharmacokinetics to Design Humanized Dosage Regimens for a Bloodstream and Lung Infection Model in Immune-Competent Mice

Yuanyuan Jiao,[a]* Jun Yan,[b] Michael Vicchiarelli,[c] Dhruvitkumar S. Sutaria,[a]§ Peggy Lu,[b] Zeferino Reyna,[b] Brad Spellberg,[d] Robert A. Bonomo,[e,f,g,h,i,j] George L. Drusano,[c] Arnold Louie,[c] Brian M. Luna,[b] Jürgen B. Bulitta[a]

[a]Department of Pharmacotherapy and Translational Research, College of Pharmacy, University of Florida, Orlando, Florida, USA
[b]Department of Molecular Microbiology and Immunology, Keck School of Medicine, University of Southern California, Los Angeles, California, USA
[c]Institute for Therapeutic Innovation, College of Medicine, University of Florida, Orlando, Florida, USA
[d]Los Angeles County-USC (LAC+USC) Medical Center, Los Angeles, California, USA
[e]Department of Molecular Biology and Microbiology, Case Western Reserve University, Cleveland, Ohio, USA
[f]Deparment of Medicine, Case Western Reserve University, Cleveland, Ohio, USA
[g]Department of Biochemistry, Case Western Reserve University, Cleveland, Ohio, USA
[h]Deparment of Pharmacology, Case Western Reserve University, Cleveland, Ohio, USA
[i]Department of Proteomics and Bioinformatics, Case Western Reserve University, Cleveland, Ohio, USA
[j]Louis Stokes Cleveland Department of Veterans Affairs Medical Center, Cleveland, Ohio, USA

ABSTRACT Polymyxin B is a "last-line-of-defense" antibiotic approved in the 1960s. However, the population pharmacokinetics (PK) of its four main components has not been reported in infected mice. We aimed to determine the PK of polymyxin B1, B1-Ile, B2, and B3 in a murine bloodstream and lung infection model of *Acinetobacter baumannii* and develop humanized dosage regimens. A linear 1-compartment model, plus an epithelial lining fluid (ELF) compartment for the lung model, best described the PK. Clearance and volume of distribution were similar among the four components. The bioavailability fractions were 72.6% for polymyxin B1, 12.0% for B1-Ile, 11.5% for B2, and 3.81% for B3 for the lung model and were similar for the bloodstream model. While the volume of distribution was comparable between both models (17.3 mL for the lung and ~27 mL for the bloodstream model), clearance was considerably smaller for the lung (2.85 mL/h) compared to that of the bloodstream model (5.59 mL/h). The total drug exposure (AUC) in ELF was high due to the saturable binding of polymyxin B presumably to bacterial lipopolysaccharides. However, the modeled unbound AUC in ELF was ~16.7% compared to the total drug AUC in plasma. The long elimination half-life (~4 h) of polymyxin B enabled humanized dosage regimens with every 12 h dosing in mice. Daily doses that optimally matched the range of drug concentrations observed in patients were 21 mg/kg for the bloodstream and 13 mg/kg for the lung model. These dosage regimens and population PK models support translational studies for polymyxin B at clinically relevant drug exposures.

KEYWORDS polymyxin B, lung epithelial lining fluid (ELF), bloodstream infection, lung infection, population pharmacokinetics, LC-MS/MS, humanized dosing, translational pharmacology, S-ADAPT, target-mediated drug disposition, lung epithelial lining fluid (ELF)

Polymyxin B was introduced to clinics in the late 1950s to 60s and currently serves as a "last-line-of-defense" antibiotic to treat serious infections by multidrug-resistant Gram-negative bacteria (1, 2). Polymyxin B is available as a mixture of structurally related polypeptides (3), with approximate proportions of 73.5% for polymyxin B1,

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Address correspondence to Jürgen B. Bulitta, jbulitta@cop.ufl.edu, or Brian M. Luna, brian.luna@usc.edu.

*Present address: Yuanyuan Jiao, Peking University Cancer Hospital, Beijing, China.

§Present address: Dhruvitkumar S. Sutaria, Genentech, CA, USA.

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8.6% for B1-Ile, 13.7% for B2, and 4.2% for B3. However, these fractions may differ to some degree between various batches of polymyxin B (4). Polymyxin B is a polycation at physiological pH, and its initial mode of action involves electrostatic binding to anionic lipopolysaccharide (LPS) binding sites on the outer membrane of Gram-negative bacteria.

Over the last 20 years, safe and effective polymyxin B dosage regimens have been developed (5–7). Although polymyxins have been used clinically for decades, contemporary bioanalytical assays have only recently been developed (3, 8–17). While several preclinical studies reported the pharmacokinetics (PK) of polymyxin E (i.e., colistin), fewer preclinical PK studies were reported for polymyxin B. Even now, many assays do not distinguish between the individual polymyxin B components (8–10).

He et al. were the first to dissect the PK of polymyxin B1, B1-Ile, B2, and B3 in uninfected (9) and infected (11) mice at a dose of 3 mg/kg. The PK of the four polymyxin B components in serum and lung epithelial lining fluid (ELF) was comparable. The same conclusion was also reported (3, 12) in uninfected rats. While these studies assessed individual polymyxin B components, they did not employ population PK modeling, did not develop humanized dosage regimens, and did not model the time course of drug concentrations in ELF. Thus, validated humanized dosage regimens in animal infection models based on population PK modeling are still needed. Humanized dosage regimens account for the PK differences (e.g., in half-life) between mice and humans. They adjust the mouse doses (including potentially smaller supplement doses) and dosing times to match the range of plasma concentrations in humans closely. The resulting PK profiles in mice are more clinically relevant. In this study, we sought to mirror a common polymyxin B dosage regimen with a loading dose of 2 mg/kg and a maintenance dose of 1.5 mg/kg every 12 h (starting at 12 h) (7).

As bacterial infection can affect the PK, there is consensus regarding determining the PK in infected animals, ideally for the infection model and bacterial strain of interest (18). For polymyxin B, studies regarding the impact of infection on PK are scarce (11, 19–21). Moreover, bacterial LPS presents an important binding site (i.e., a reservoir), which may considerably contribute to the total polymyxin concentration in ELF (10, 22). Population PK modeling is the most powerful approach to analyze sparse data obtained with destructive sampling (i.e., one observation time [e.g., for plasma and ELF] per animal) (23, 24), while accounting for between-subject variability.

This study aimed to determine the population PK of the four main polymyxin B components in a murine bloodstream and lung infection model. This allowed us to compare the PK between both infection models and characterize the drug exposures in ELF. Second, we sought to define and validate humanized dosage regimens in mice. Highly sensitive and specific ultraperformance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) assays were developed and allowed us to reliably distinguish between the four polymyxin B components. These population PK models and humanized dosage regimens were developed based on a series of PK dose range, PK validation, and efficacy studies in infected, immunocompetent mice. The developed population PK models and humanized dosage regimens serve as translational tools for nonclinical evaluations of polymyxin B at clinically relevant drug exposures.

RESULTS

Mice showed multiple signs of infections as previously published in our natural history study (25). The lower limit of quantification (LLOQ) for all analytes was 0.025 mg/L in plasma and 0.00625 mg/L in BAL fluid (see supplemental material for additional assay results). The population PK analyses sought to develop robust models for the four polymyxin B components using destructive sampling data sets. We eventually fixed the parameters of the residual error model and showed via sensitivity analyses that this choice had a limited impact on the population PK parameter estimates (see supplemental material). Our Monte Carlo simulation estimation studies further showed

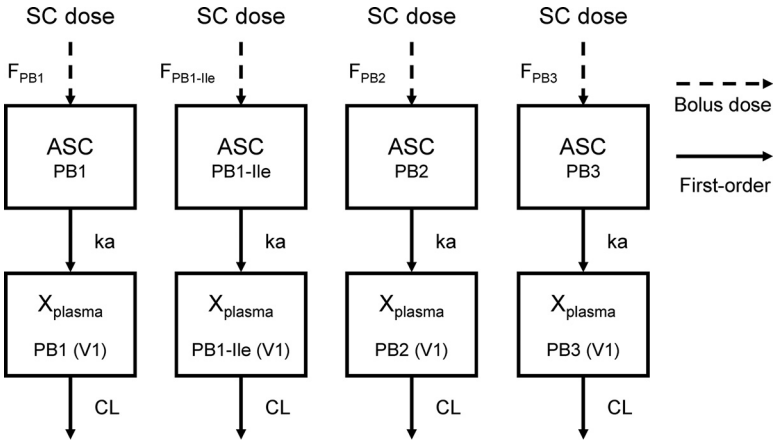


FIG 1 Population PK model structure for polymyxin B1, B1-Ile, B2, and B3 in plasma for the murine bloodstream infection model. Clearance (CL) and volume of distribution (V1) and the absorption half-life after subcutaneous (SC) dosing were shared between the four components. The total bioavailability was assumed to be 1. However, the fractional bioavailability was estimated for each of the components (e.g., F_{PB1}) with the sum of four bioavailability terms being 100%. The X_{plasma} represents the amount of the respective compound in the central compartment, and ASC is the amount of drug in the subcutaneous compartment.

that both plasma and ELF parameters could be estimated without bias and with adequate precision (see Table S1 in the supplemental material).

Bioavailability. The population PK model for the bloodstream model included first-order absorption and one disposition compartment (Fig. 1), and the lung infection model contained an additional ELF compartment (Fig. 2). The absorption half-life was initially estimated to be similar at the low and intermediate doses (i.e., 8 and 12 mg/kg), and this estimate was shared in the final model. At the high dose (16 mg/kg), absorption was slower. This model yielded unbiased and reasonably precise individual and population-fitted concentrations (Fig. S2, S3, and S4). A four-way split (26) was implemented to describe the relative bioavailability for each of the four polymyxin B components with the total bioavailability (F) being 100% for the sum of all components (Fig. S5 and S6). The estimated bioa-

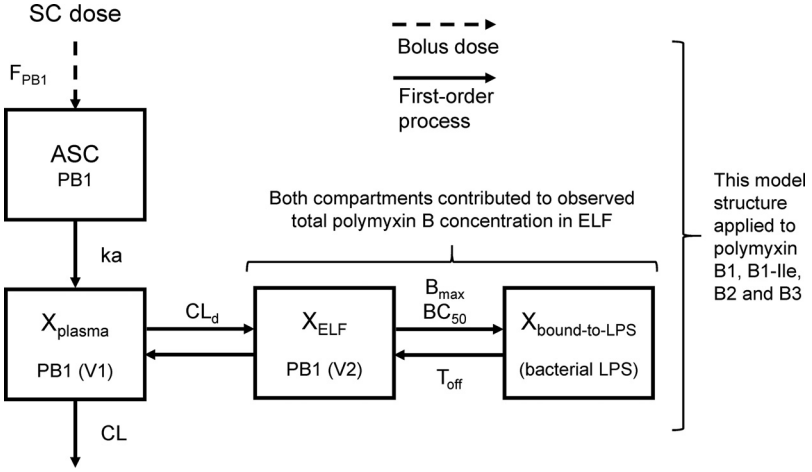


FIG 2 Population PK model structure for polymyxin B1, B1-Ile, B2, and B3 in plasma and lung ELF for murine lung infection model. The model structure is shown for polymyxin B1, and the same structure with separate compartments was applied for the other three components. The estimates for clearance (CL), volume of distribution of the central compartment (V1), and the absorption half-life after subcutaneous (SC) dosing were shared between the four components. The amount ($X_{bound-to-LPS}$) represents polymyxin B bound to LPS in the lung. While the binding of polymyxin B1 to LPS is shown, all four components were modeled to compete at the same binding sites with a shared maximum binding capacity (B_{max}), shared amount of polymyxin B associated with half-maximal binding (BC_{50}), and dissociation half-life (T_{off}; see model code in the supplemental material).

TABLE 1 Population PK parameter estimates for the model with different CL and different V1 between the four studies in the murine bloodstream infection model[a]

Symbol (unit)	Parameter	Population mean, CV[b] (SE%)	Between subject variability, CV[b] (SE%)
$T1/2_{abs}$ (L) (min)	Absorption half-life at 8 and 12 mg/kg	7.30 (10.0%)	0.147 (59.5%)
$T1/2_{abs}$ (H) (min)	Absorption half-life at 16 mg/kg	11.9 (15.9%)	0.176 (83.8%)
CL1 (mL/h)	Clearance for study 1 (dose range study)	5.59 (5.1%)	0.364 (26.8%)
CL2 (mL/h)	CL for study 2 (dose regimen validation study)	4.56 (4.4%)	0.0672 (75.9%)
CL3 (mL/h)	CL for study 3 (HUMC1 strain)	7.55 (11.2%)	0.356 (50.9%)
CL4 (mL/h)	CL for study 4 (LAC-4 Col^R strain)	0.501 (22.7%)	0.247 (97.1%)
V1 study 1 (mL)	Volume of distribution for the central compartment for study 1 (dose range study)	27.3 (4.7%)	0.328 (24.1%)
V1 study 2 (mL)	Volume for study 2 (dosage regimen validation study)	26.3 (9.8%)	0.455 (27.6%)
V1 study 3 (mL)	Volume for study 3 and the HUMC1 strain	38.8 (13.3%)	0.395 (47.9%)
V1 study 4 (mL)	Volume for study 4 and the LAC4 strain	25.9 (14.5%)	0.609 (34.1%)
F_{PB1}	Fractional bioavailability of PB1	72.8% (2.0%)	Small[c]
$F_{PB1\text{-}Ile}$	Fractional bioavailability of PB1-Ile	11.7% (1.9%)	Small[c]
F_{PB2}	Fractional bioavailability of PB2	12.6% (2.1%)	Small[c]
F_{PB3}	Fractional bioavailability of PB3	2.91%	Small[c]

[a]This model shared CL and V1 among the four polymyxin B components in each study. The sum of bioavailability after subcutaneous administration for the four components of polymyxin B was assumed to be 100%. All additive residual error standard deviations were fixed to 0.025 mg/L (equivalent to 1× the LLOQ in plasma). All proportional residual errors were fixed to a coefficient of variation of 10% for plasma.
[b]Coefficient of variation (CV%).
[c]On logistically transformed scale, the standard deviations were 0.200 for TF_B1, 0.127 for DF_B1ILE, and 0.378 for DF_B2. Please see model code (Fig. S5) for the implementation of these parameters.

vailability fractions were consistent across the four components between all models and data sets. Polymyxin B1 was the predominant component with a bioavailability fraction of ~73%, B1-Ile and B2 each had bioavailability fractions of 11.5 to 12.6%, and B3 accounted for 3% to 4% (Table 1 and 2).

ELF penetration. The ELF compartment in the population PK model for lung infection had a small fixed volume (2 mL) and contained a saturable binding site (Fig. 2 and

TABLE 2 Population PK parameter estimates for polymyxin B1, B1-Ile, B2, and B3 in murine lung infection model[a]

Symbol (unit)	Parameter	Population mean, CV (SE%)[c]	Between subject variability, CV (SE%)[c]
$T1/2_{abs}$ (min)	Absorption half-life	9.0 (fixed)	0.20 (fixed)
CL1 (mL/h)	Clearance for study 1 (dose range study)	2.85 (5.0%)	0.291 (29.6%)
CL2 (mL/h)	CL for study 2 (dose regimen validation study)	2.54 (13.2%)	0.168 (145%)
CL3 (mL/h)	CL for study 3 and the HUMC1 strain	6.89 (9.6%)	0.292 (36.7%)
CL4 (mL/h)	CL for study 4 and the LAC-4 Col^R strain	6.55 (10.0%)	0.266 (54.7%)
V1 (mL)	Volume of distribution for the central compartment	15.3 (4.7%)	0.345 (24.8%)
V2 (mL)	Volume of distribution for the ELF compartment	2.00 (fixed)	0.15 (fixed)
CLd (mL/h)	Distribution clearance to the ELF compartment	11.5 (26.6%)	0.15 (fixed)
F_{ELF}	Ratio of polymyxin B penetrated into ELF	0.167 (6.1%)	0.4 (fixed)
B_{max08} (1/h)	Maximum binding rate constant (B_{max}) to LPS in the ELF compartment at 8 mg/kg dose (study 1)	16.3 (13.1%)	0.20 (fixed)
B_{max12} (1/h)	B_{max} at 12 mg/kg dose (study 1)	11.0 (22.4%)	0.20 (fixed)
B_{max16} (1/h)	B_{max} at 16 mg/kg dose (study 1)	37.4 (16.6%)	0.20 (fixed)
B_{max2} (1/h)	B_{max} in study 2	29.7 (19.8%)	0.20 (fixed)
B_{max3} (1/h)	B_{max} in study 3	10.7 (22.2%)	0.20 (fixed)
B_{max4} (1/h)	B_{max} in study 4	8.09 (28.0%)	0.20 (fixed)
BC_{50} (μg)	Amount of unbound polymyxin B (i.e. sum of the four components) in ELF resulting in 50% binding	0.1 (fixed)	0 (fixed)
T_{off} (h)	Dissociation half-life from LPS-binding in ELF	1.49 (17.2%)	0.15 (fixed)
F_{PB1}	Fractional bioavailability of PB1[b]	72.6% (1.8%)	Small[d]
$F_{PB1\text{-}Ile}$	Fractional bioavailability of PB1-Ile[b]	12.0% (2.3%)	Small[d]
F_{PB2}	Fractional bioavailability of PB2[b]	11.5% (1.6%)	Small[d]
F_{PB3}	Fractional bioavailability of PB3[b]	3.81%	Small[d]

[a]This model assumed the same clearance and volume of distribution for the four components of polymyxin B. All additive residual error standard deviations were fixed to 0.025 mg/L (equivalent to 1× the LLOQ in plasma). Proportional residual errors were fixed to a CV of 10% for plasma and to a CV of 20% for ELF.
[b]The sum of bioavailability after subcutaneous administration for the four components of polymyxin B was assumed to be 100%.
[c]Coefficient of variation (CV%).
[d]On logistically transformed scale, the standard deviations were 0.2 (fixed) for TF_B1 and DF_B1ILE, as well as 0.246 for DF_B2. Please see model code (Fig. S6) for the implementation of these parameters.

TABLE 3 Simulated unbound and total ELF to plasma ratio for various time intervals in immunocompetent mice with lung infection treated with 8 mg/kg polymyxin B subcutaneously[a]

Time interval (h)	$AUC_{unbound,ELF}$ (mg·h/L)	$AUC_{total,ELF}$ (mg·h/L)	$AUC_{total,\ plasma}$ (mg·h/L)	$AUC_{unbound,ELF}$/ $AUC_{total,\ ELF}$	$AUC_{unbound,ELF}$/ $AUC_{total,\ plasma}$	$AUC_{total,ELF}$/ $AUC_{total,\ plasma}$
0–6	6.02	17.5	39.5	0.344	0.152	0.442
0–12	8.33	36.6	53.1	0.227	0.157	0.690
0–24	9.65	70.6	60.6	0.137	0.159	1.17
0–48	10.3	126	64.0	0.082	0.161	1.96
0–168	11.3	295	68.8	0.038	0.164	4.28
0–Infinity	12.2	557	73.1	0.022	0.167	7.62

[a]These simulations were based on the established population PK model using a typical B_{max} of 16.3 h^{-1} and a single dose of 8 mg/kg. The unbound and total ELF to plasma ratios were the same across each individual component for polymyxin B, therefore one value for the sum of all four components was reported.

Table 2). This volume was chosen since it had a limited impact on the plasma PK and since it supported robust estimation of the saturable binding of polymyxin B in ELF (Table S1). The total drug area under the curve (AUC) in ELF was similar to or larger than the total drug AUC in plasma. Thus, initial models fixed the F_{ELF} to 1.0 and used a saturable binding in the lung to further allow for larger total (i.e., LPS-bound plus unbound) polymyxin B AUCs in ELF. At this stage, the dissociation half-life from the LPS binding site (T_{off}) was estimated to be large and eventually fixed to 100 h (Table S1). This saturable binding site represented a presumed binding of polymyxin B to bacterial outer membrane components (i.e., LPS) in the lung. At this stage, B_{max} showed considerable variability between data sets and the amount of unbound polymyxin B (i.e. sum of the four components) resulting in 50% of B_{max} (BC_{50}) estimate was higher (Table S1) than that of the final model (Table 2).

To distinguish between total and unbound polymyxin B in ELF, we estimated the F_{ELF} (Fig. S6) and performed a thorough sensitivity analysis for T_{off}. Estimating F_{ELF} yielded more consistent B_{max} values between the four studies and a shorter T_{off} (1.49 h), as well as smaller BC_{50} estimates (Table 2); the latter agreed well with published binding data of polymyxins to bacterial LPS (22). Thus, BC_{50} was eventually fixed to a literature estimate for polymyxin binding to bacterial LPS. Models with estimated F_{ELF} provided superior objective functions and predictive performance compared to earlier models with F_{ELF} fixed to 1. The F_{ELF} of 0.167 (6.1% relative standard error [RSE]; Table 2) was comparable to the unbound fraction of polymyxin B in mouse plasma (f_u, 0.086) (19), whereas the ratio of the total drug AUC in ELF and total drug AUC in plasma was larger. To illustrate the drug exposures over relevant time intervals, we simulated the unbound and total drug AUC in ELF and the total drug AUC in plasma (Table 3 and Fig. 3). The ratio of the unbound AUC in ELF and the total drug AUC in plasma was similar across all time intervals (i.e., 0.152 to 0.167). However, the ratio of the total drug AUC in ELF divided by that in plasma was considerably larger ($\geq$0.442) and increased over time (Table 3).

The maximum binding capacity in the lung (i.e. the rate constants, B_{max}) showed some variability with estimates ranging from 8.09 to 37.4 h^{-1} for the different doses and studies (Table 2). Including saturable binding in ELF was necessary to distinguish between the modeled unbound and observed total polymyxin B concentrations in ELF. Of note, across all lung binding models evaluated, all clearance, volume of distribution, and relative bioavailability estimates were robust and consistent. This included the data set that lacked the ELF data. Clearance and B_{max} could be estimated simultaneously, and their joint uncertainty was small (i.e., r within -0.58 and $+0.55$). The B_{max} allowed the model to have a slower decline of ELF concentrations and larger total drug AUCs in ELF compared to those in plasma.

Disposition. When estimating the clearances and volumes of distribution separately for the four components, the individual estimates were well comparable. Following the rule of parsimony, we shared these clearance and volume estimates among the four components and the simplified models still provided satisfactory curve fits.

The polymyxin B clearance in the different studies ranged from 2.54 to 7.55 mL/h for most data sets (Table 1 and 2). In the dose range studies (study 1), clearance was approximately 2-fold larger for the bloodstream (5.59 mL/h; RSE, 5.1%) compared to that in the lung infection model (2.85 mL/h; RSE, 5.0%). Of note, the disease severity

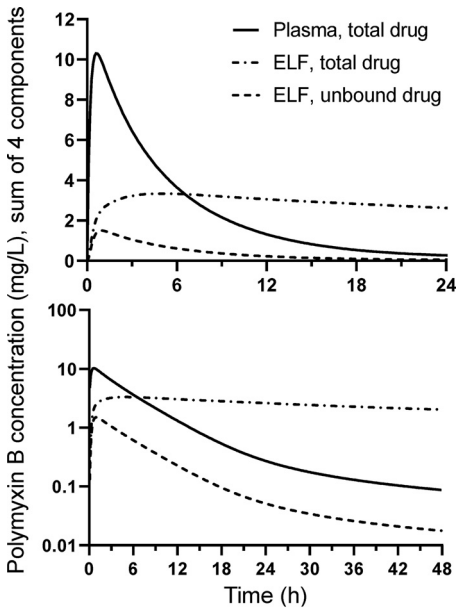


FIG 3 Predicted concentration-time profiles for the total polymyxin B concentration in plasma and ELF, as well as the modeled unbound polymyxin B concentration in ELF on linear (top) and log (bottom) scale. While the unbound ELF concentrations paralleled the total drug concentrations in plasma, the total drug concentration in ELF (i.e., bound plus unbound) rose to higher levels and declined more slowly. This simulation applies to a single dose of 8 mg/kg and a B_{max} of 16.3 h^{-1} (as in dose range study 1 for the lung model in Table 2 and as used in Table 3), with a dissociation half-life (T_{off}) of 1.49 h from the saturable binding site in the lung.

was greater for mice in the bloodstream compared to mice in the lung infection model. This likely caused the clearance to be small (0.501 mL/h) in study 4 for the bloodstream model (Table 1). The volume of distribution ranged from 25.9 to 38.8 mL among the four studies in the bloodstream model (Table 1). In all four lung model studies, the volume of distribution at steady state was 17.3 mL (i.e., the sum of both volumes; Table 2) and was slightly smaller than the volume of distribution for the bloodstream model.

The terminal half-lives $t_{1/2}$ ranged from 3.39 to 4.74 h for most data sets. In the dose range studies (study 1), terminal half-life was 4.22 h for the lung and 3.39 h for the bloodstream model. These relatively long half-lives supported dosing of mice every 12 h for humanized regimens. The terminal half-life in study 4 (with strain LAC-4 COL^R) was 35.8 h, likely since mice were very sick and clearance vanished. Visual predictive checks showed that these models reasonably captured the central tendency and variability for the different doses and studies of the bloodstream (Fig. 4) and lung infection models (Fig. 5 and 6). The underpredictions for polymyxin B3 in ELF, but not in plasma, in studies 2 to 4 may have been caused by differences in viable counts and LPS production in various studies.

Humanized dosage regimens. We considered humanized dosage regimens for the bloodstream infection model with either two doses per day (i.e., 11 mg/kg at 0 h and 10 mg/kg at 12 h) or four doses per day (i.e., 9.5 mg/kg at 0 h, 1.5 mg/kg at 6 h, 8.5 mg/kg at 12 h, and 1.5 mg/kg at 18 h; Fig. 7 and Fig. S7). The doses at 6 and 18 h were supplement doses to offset the slightly shorter half-life in mice compared to that in patients. The simpler two-dose regimen was selected due to its better feasibility. The same process was applied to the lung infection model, and a humanized dosage regimen with two doses per day was selected (i.e., 7 mg/kg at 0 h and 6 mg/kg at 12 h; Fig. 7 and Fig. S7).

DISCUSSION

This study found similar or higher total drug exposures (i.e., AUC) in ELF compared to those in plasma, and the ratio of total drug AUCs increased over time (Table 3). Population PK modeling explained these high drug exposures and the slow decline of

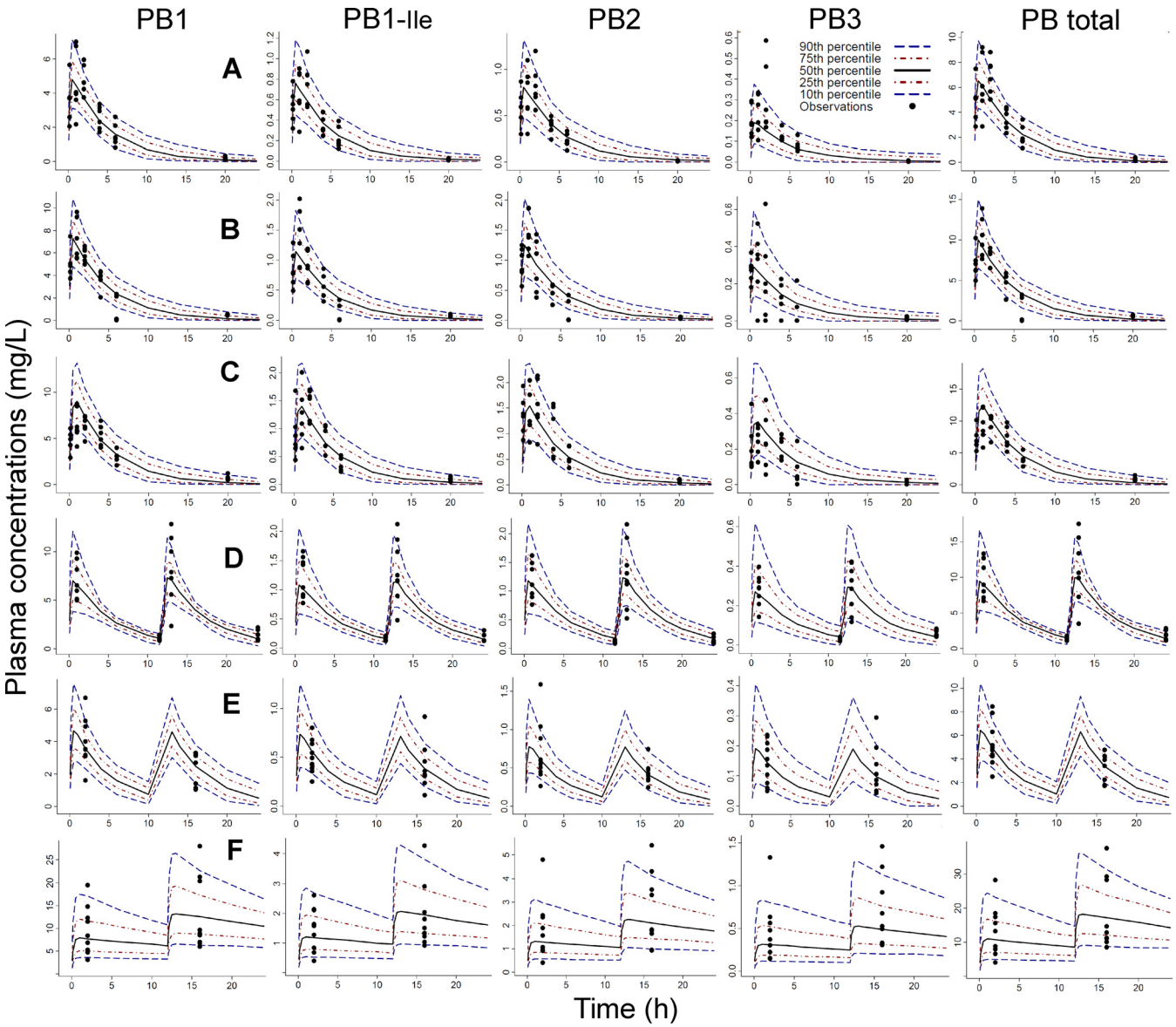


FIG 4 Visual predictive checks for plasma concentrations of polymyxin B1, B1-Ile, B2, B3, and total (i.e. sum of the four components) polymyxin B (PB) in the murine bloodstream infection model at various dose levels from all substudies. Rows A, B, and C show the results for study 1 (8, 12, and 16 mg, single dose), whereas rows D to F apply to studies 2 to 4, which used a humanized dosage regimen (i.e., 11 mg/kg at 0 h and 10 mg/kg at 12 h). The plots show the observations (markers), the 50th percentile (i.e., the median) of the model predictions (black line) along with the 80% prediction interval [10th to 90th percentile], and the interquartile range [25th to 75th percentile]. Ideally, the median should capture the central tendency of the observations and 10% of the observations should fall outside the 80% prediction interval on either side.

ELF concentrations by saturable binding of polymyxin B in the lung presumably to bacterial LPS (Fig. 3). This presents a target-mediated drug disposition scenario (as reported for other small molecules; references 27–32) with bacterial LPS presumably serving as a high-affinity binding site in the peripheral compartment (i.e., ELF). Such LPS may be continuously synthesized by viable bacteria and further arise from outer membrane fragments of killed bacteria. Moreover, outer membrane vesicles may present additional decoy targets (33–35). The impact of LPS binding on the PK of polymyxin B is likely affected by the rate of penetration between plasma and ELF as well as by the residence time of LPS (as part of membrane components) in the lung. The modeled unbound AUC in ELF was 16.7% of the total drug AUC in plasma, roughly in agreement with the unbound fraction of polymyxin B in mouse plasma (19). Future studies are needed that ideally simultaneously determine the PK in plasma and ELF, bacterial load, and efficacy in mice. These studies should identify the

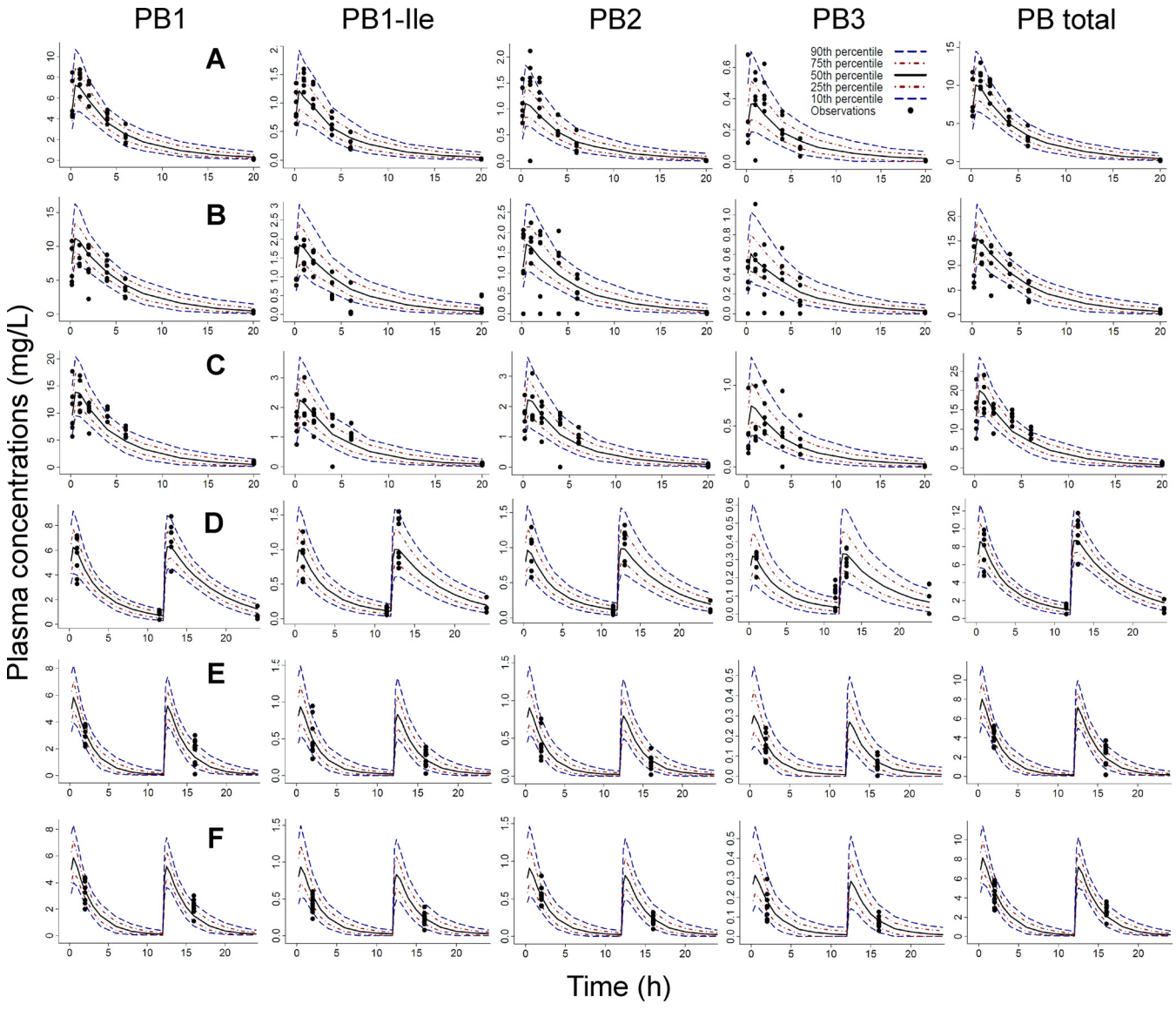


FIG 5 Visual predictive checks for plasma concentrations of polymyxin B1, B1-Ile, B2, B3, and total (i.e. sum of the four components) polymyxin B in the murine lung infection model at various dose levels from all substudies. Rows A, B, and C show the results for study 1 (8, 12, and 16 mg, single dose), whereas rows D to F apply to studies 2 to 4, which used a humanized dosage regimen (i.e., 7 mg/kg at 0 h and 6 mg/kg at 12 h). Please see Fig. 4 for further explanations on the interpretation of visual predictive checks.

fraction of the drug exposure in ELF that is pharmacologically active (10). Further, this is the first study to prospectively validate humanized dosage regimens for polymyxin B in two immunocompetent mouse infection models.

We used population PK to simultaneously model four polymyxin B components and showed that they had similar clearances and volumes of distribution, in agreement with prior studies (9, 11). Across all data sets and models, the fractional bioavailability estimates consistently showed that polymyxin B1 comprised the majority of polymyxin B, followed by polymyxin B1-Ile and polymyxin B2, and by polymyxin B3 as minor components. The observed differences in the PK parameters between the two infection models (Table 1 and 2) may be attributable to alterations in vascular penetration and organ functions induced by inflammation in severe infections (25). By estimating the lung PK data with and without the ELF concentrations across each stage (Fig. S1), we confirmed this clearance difference was not caused by the saturable binding in the lung.

Inclusion of saturable binding (10, 22) for the four polymyxin B components to the same binding site in lung ELF, presumably LPS from *A. baumannii*, was required to fit the

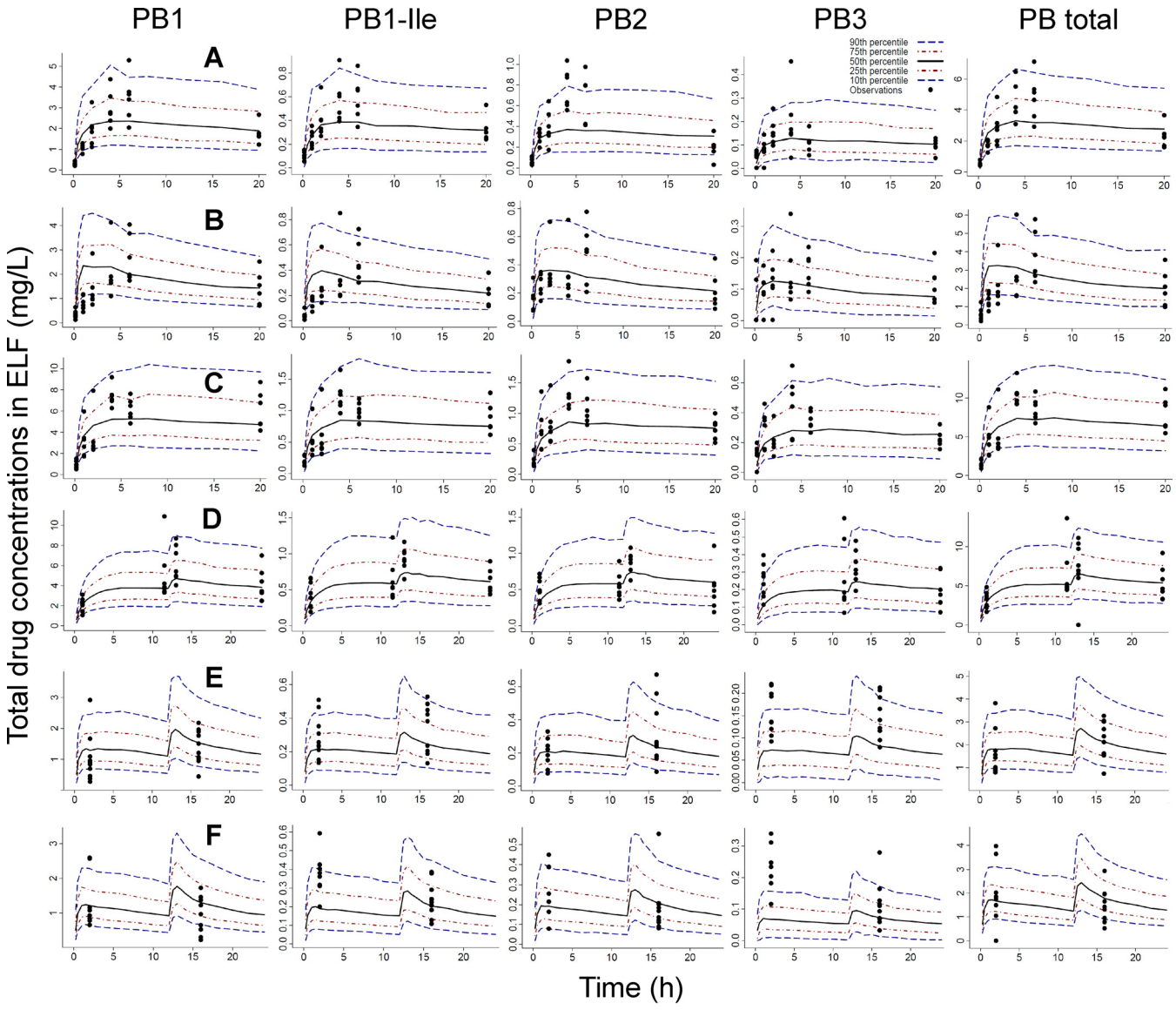


FIG 6 Visual predictive checks for the total ELF concentrations (i.e. bound plus unbound) of polymyxin B1, B1-Ile, B2, B3, and total (i.e. sum of the four components) polymyxin B in the murine lung infection model at various dose levels from all substudies. Rows A, B, and C show the results for study 1 (8, 12, and 16 mg, single dose), whereas rows D to F apply to studies 2 to 4, which used a humanized dosage regimen (i.e., 7 mg/kg at 0 h and 6 mg/kg at 12 h). Please see Fig. 4 for further explanations on the interpretation of visual predictive checks.

slow decline of the total (i.e., bound plus unbound) polymyxin B concentrations in ELF (Fig. 3, 5, and 6). The estimated B_{max} differed to some degree between strains and studies (Table 2), which might be explained by differences in the amount of LPS and in the bacterial counts in ELF. The estimate for BC_{50} was close to literature values for the binding of polymyxins to bacterial LPS (22, 36, 37) and thus eventually fixed to 0.1 μg. This supported the estimation of the other binding parameters. In simulations, the unbound polymyxin B concentrations in ELF paralleled the total drug concentrations in plasma; in contrast, the total drug concentrations in ELF were higher and declined slowly (Fig. 3). Thus, the total drug AUC in ELF divided by that in plasma increased from 0.442 for the $AUC_{0–6h}$, to 1.17 for the $AUC_{0–24h}$, and to 7.62 for the $AUC_{0\text{-infinity}}$. These total drug ELF/plasma AUC ratios were in the range of prior estimates of 0.59 for the $AUC_{0\text{-infinity}}$ in noninfected mice (n = 4) (9) and 0.57 for the $AUC_{0–6h}$ in mice infected by *Pseudomonas aeruginosa* for polymyxin B1 (11). Overall, accounting for the saturable binding in ELF was important to describe the PK of polymyxin B in ELF and our estimated penetration ratio (F_{ELF}) was comparable to the unbound fraction of polymyxin B in mouse plasma (19).

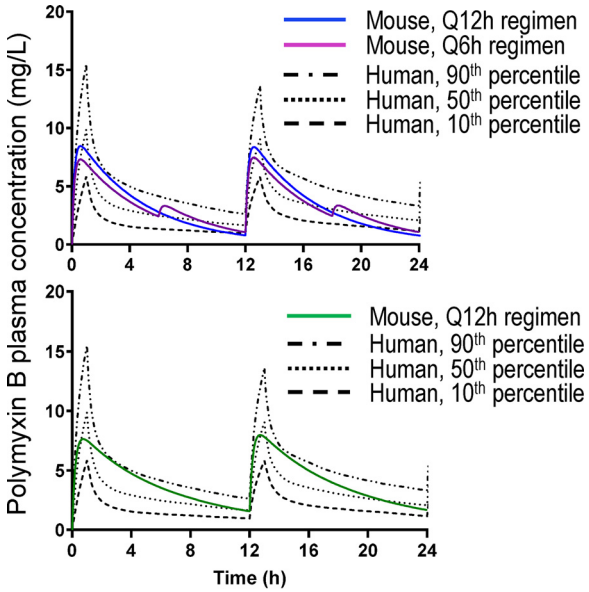


FIG 7 Humanized dosage regimens for polymyxin B in murine bloodstream (top) or lung (bottom) infection models. Both regimens used the 2 mg/kg loading dose with a 1.5 mg/kg q12h maintenance dose for polymyxin B in critically ill patients as clinical reference. The dashed lines are the simulated 10th, 50th (median), and 90th percentiles of polymyxin B PK profiles in human plasma. In the plots for the bloodstream infection model, the blue line indicates the simulated plasma PK profile in mice for a Q12h dosage regimen (11 mg/kg at 0 h and 10 mg/kg at 12 h), while the purple line is for a Q6h dosage regimen (9.5 mg/kg at 0 h, 1.5 mg/kg at 6 h, 8.5 mg/kg at 12 h, and 1.5 mg/kg at 18 h). In the plots for the lung infection model, the green line is the simulated total plasma PK profile in mice for a Q12h dosage regimen (7 mg/kg at 0 h and 6 mg/kg at 12 h).

The PK of polymyxin B components in animals has been reported previously (3, 9, 11, 12). He et al. (9) showed that the four polymyxin B components have similar clearances (range, 21.4 to 25.4 mL/h; for mice with 22 g weight) and slightly more different volumes of distribution (7.7 to 13.3 mL) in noninfected mice. In a neutropenic murine pneumonia model challenged with resistant *P. aeruginosa*, He et al. (11) also reported comparable serum PK for the four polymyxin B components with clearances ranging from 6.8 to 12 mL/h (for 22 g mice). The same group reported an even smaller clearance of 2.75 mL/h in a neutropenic murine pneumonia model with *A. baumannii* (21). Taken together, clearances were smaller in infected compared to noninfected mice. Our clearance estimates for polymyxin B (Table 1 and 2) were within the range of those previously reported for mice infected with *A. baumannii* or *P. aeruginosa*.

Landersdorfer et al. (19) employed population modeling to assess the PK of total polymyxin B (i.e., without separating the components) in a neutropenic mouse thigh infection model challenged with 10^5 to 10^6 *K. pneumoniae* over a wide dose range (2 to 32 mg/kg). The authors identified saturable absorption and a linear plus saturable elimination (19), and clearance was most noticeably saturated at the two highest doses (16 and 32 mg/kg). With exception of a single dose at 16 mg/kg in study 1, all of our polymyxin B doses were at or below 12 mg/kg. These smaller doses likely allowed us to use a first-order absorption model (Table 1 and 2). Further, a linear clearance adequately described our PK data likely due to our lower doses. The inclusion of a saturable clearance did not improve the models (results not shown). Our clearance estimates (Table 1 and 2) were well comparable to those reported by Landersdorfer et al. (19), who reported a mean clearance of 10.7 mL/h (in 22 g mice) at low total polymyxin B concentrations and of 4.2 mL/h at 20 mg/L polymyxin B. Likewise, our estimated volumes of distribution at steady state agreed well with previously reported values of 24.9 mL (for 22 g mice) (19).

Tam et al. (38) found no significant difference in the *in vitro* potencies of various polymyxin B components against several clinical isolates of multidrug-resistant bacteria. With

the free-plasma AUC (*f*AUC)/MIC of polymyxins best-predicting efficacy in mice (19), matching the *f*AUC/MIC for the sum of the four polymyxin B components seemed plausible. Moreover, polymyxins are known to cause rapid killing at high concentrations *in vitro* (22), suggesting that peak concentrations in mice should not exceed those in patients (Fig. 7).

The present study developed and prospectively validated the first humanized dosage regimens for polymyxin B using two *A. baumannii* infection models in immunocompetent mice. Of note, the higher clearance in the bloodstream compared to that in the lung infection model led to an approximately 62% higher daily dose (21 mg/kg for the bloodstream versus 13 mg/kg for the lung model; Fig. 7) in our humanized dosage regimen. We attempted to match the peak and trough concentrations between mice and humans, with concentrations in mice staying within the 10th to 90th percentiles of concentrations in critically ill patients. Due to optimizing all three criteria, the ratio of daily doses (21 versus 13 mg/kg) was slightly smaller than the 2-fold ratio in clearances between the bloodstream and lung infection models (Table 1 and 2). Developing humanized dosage regimens (18, 39) was further complicated by the biexponential decline of plasma concentrations in humans compared to a (largely) monoexponential decline in mice (Fig. S7). This scenario was manageable for polymyxin B, as the α-phase only contributed 11% of the total drug AUC in human plasma, whereas the β-phase comprised 89% of the AUC in plasma (40).

The overall relatively long elimination half-life of polymyxin B in mice allowed us to develop humanized dosage regimens for the lung and bloodstream infection models with doses at 0 and 12 h. This study showed that the attainment of the targeted drug exposures should ideally be confirmed using the same infection model (including the same strain of mice), infecting bacterial strain(s) (resources permitting), and bacterial inoculum as those used in the efficacy studies. For the lung model, this was emphasized by the target-mediated drug disposition due to the saturable binding of polymyxin B in ELF.

One limitation of the present study is that we did not quantify the unbound concentrations of polymyxin B in mouse plasma and instead assumed a similar unbound fraction in mice and patients based on prior studies (19, 41, 42). Moreover, the unbound drug concentrations in ELF arose from our mathematical model and were not experimentally quantified. As a further potential limitation, the present study did not identify the saturable elimination of polymyxin B in mice at doses up to 16 mg/kg. Moreover, we used a small, fixed value (2 mL) for the volume of distribution of the ELF compartment (Fig. 2) to support the estimation of saturable binding in lung ELF (Table S1). However, future studies should consider evaluating different volumes for the ELF compartment. Finally, the present study assessed one polymyxin B susceptible and one resistant isolate. Studying more isolates with several PK sampling times would allow one to draw more robust conclusions about the impact of bacterial strains on the PK (43).

In summary, this study showed that the PK of polymyxin B was adequately described by a linear model at doses up to 16 mg/kg. The polymyxin B clearance was approximately 2-fold smaller in the lung compared to that in the bloodstream infection model for mice challenged with two *A. baumannii* strains. The validated humanized dosage regimens for polymyxin B used a 62% higher daily dose in the bloodstream (21 mg/kg) compared to that in the lung infection model (13 mg/kg). The long half-lives of polymyxin B in infected mice (3.39 to 4.74 h) allowed us to dose polymyxin B every 12 h in mice for humanized dosage regimens. Our population PK models simultaneously fit the four components of polymyxin B across four studies for each infection model. This is the first population model to identify the saturable binding of polymyxin B presumably to bacterial LPS in the lung, which was essential to describe the slow decline of total drug concentrations in ELF. This proposed binding site presented a "sink" for polymyxin B. In the model, the estimated unbound drug exposure in ELF was 16.7% compared to the total drug exposure in plasma. Future research is warranted to identify the pharmacologically active fraction of the observed total polymyxin B exposure in ELF. Such studies should quantify both active (i.e., unbound) polymyxin B concentrations (e.g., via a bioassay) and total polymyxin B (via UPLC-MS/MS) in BAL fluid and thus in ELF. This study enhanced our understanding of the

PK for the individual polymyxin B components in two mouse infection models. The developed humanized dosage regimens and population PK models will benefit further translational PK/PD studies for polymyxin B.

MATERIALS AND METHODS

The present work included four sequentially performed studies, with a PK dose range study (study 1), a PK validation study (study 2), and two efficacy studies (studies 3 and 4) with mice infected by two *A. baumannii* strains. Two immunocompetent murine infection models were employed for all studies. These were a bloodstream infection model challenged via the intravenous (IV) route and a lung infection model challenged by oral aspiration (OA). The infecting bacterial strains (HUMC1 and LAC-4) were originally isolated from patients in Los Angeles County, CA (44). The LAC-4 Col^R strain was a spontaneous colistin-resistant mutant. The polymyxin B MIC was 0.25 mg/L for the susceptible strain HUMC1 and 64 mg/L for the resistant strain LAC-4 Col^R. Studies 1, 2, and 3 used strain HUMC1, and study 4 used LAC-4 Col^R.

In-house PK data for polymyxin B were used to identify informative sampling times for study 1. Population PK models were developed for the data of each PK study. With study 1 providing frequent PK samples at three dose levels, the model structure was kept and parameters reestimated upon the availability of additional PK data from studies 2 to 4. In the final analysis, the PK data from all studies were combined in one data set and simultaneously estimated. Throughout this 'learn-and-confirm' process (45), a robust understanding was gained of the PK of polymyxin B in both infection models (Fig. S1).

All animal work was conducted following approval by the Institutional Animal Care and Use Committee (IACUC protocol 20810) at the University of Southern California, in compliance with the recommendations in the *Guide for the Care and Use of Laboratory Animals* of the National Institutes of Health. Mice were monitored at least twice daily for 7 days. Weight loss of greater than 15% of body weight triggered euthanasia. Soft bedding and other enrichment devices were provided as recommended by the vet staff. Nutritional supplements, such as the hydrogel packs, were provided as needed.

Polymyxin B for animal PK studies was purchased from Novaplus (Lake Zurich, IL). Polymyxin B1, isoleucine B1 (B1-Ile), B2, and B3 reference standards used in bioanalysis were from TOKU-E (Bellingham, WA). Urea (purity >99%) and its stable isotope-labeled internal standard urea-$^{15}N_2$ (purity >98%) were purchased from Sigma-Aldrich (St. Louis, MO). Acetonitrile, methanol, formic acid, phosphoric acid, and all the other reagents used in bioanalysis were LC/MS grade purchased from Fisher Scientific.

Specific pathogen-free, healthy male (8 weeks) and female (10 weeks), immunocompetent C3HeB/FeJ mice (Jackson Laboratory; stock no. 000658) were used, as this breed has been shown to be more susceptible to infection caused by *A. baumannii* than other commonly used inbred mouse strains (46). Female mice were slightly older to achieve similar body weights. Food and water were available to mice *ad libitum*.

Two murine infection models were established, a bloodstream infection model challenged by the IV route and a lung infection model challenged by the OA route, as reported previously (25). Initial bacterial inocula ranged from 10^7 CFU to $5{\cdot}10^8$ CFU (see supplemental material for details). Polymyxin B solutions were freshly prepared in 0.9% saline and administered subcutaneously starting at 2 h postinfection. For study 1, each dose group included 36 animals with equally distributed sexes (weighed before dosing). Drug concentrations in the dosing solution (300 μL) were adjusted based on weight. Single doses of 8, 12, or 16 mg/kg were used for the dose range study (study 1). Humanized regimens were assessed for the PK validation (study 2) and efficacy studies (studies 3 and 4).

Optimal doses and sampling times were determined using the PopED lite software (47) using a population PK model for total polymyxin B developed from our in-house data set. This optimal design was validated via a parametric Monte Carlo simulation estimation approach (see below) using a design with one sampling time (i.e., for plasma and ELF) per mouse. The optimal sampling times for study 1 were 0.17, 1, 2, 4, 6, and 20 h. The observation times in the PK validation study (study 2) were chosen based on these informative sampling times and on the pharmacologically important times for antibacterial efficacy (i.e., peak and trough concentrations). The PK validation study (study 2) used sampling times of 1, 11.5, 13, and 24 h. In efficacy studies 3 and 4, a reduced set of PK samples was obtained (at 2 and 16 h postdosing) to confirm the average drug exposure.

At the scheduled PK sampling times, animals were anesthetized intraperitoneally with 300 μL of a 30 mg/mL ketamine and 3 mg/mL xylazine solution and heparinized intraperitoneally with 20 to 30 μL (i.e., 200 to 300 units USP). For all groups, at least 500 μL of blood was collected at the scheduled time. For the lung infection model, bronchoalveolar lavage (BAL) fluid samples were collected using 1 mL of PBS for lavage via the trachea. All PK samples were placed on ice immediately after collection, centrifuged for 10 min at 4°C, and aliquoted. All samples were stored at −70°C or below before analysis and shipped on dry ice to the University of Florida for bioanalysis.

Bioanalysis. After thawing, plasma and BAL fluid samples were vortexed for 30 s, and analyzed by validated UPLC-MS/MS methods (10). Samples (20 μL for plasma and 50 μL for BAL fluid) were spiked with 5 μL of the internal standard avibactam (250 mg/L in H_2O). Then, a 4% H_3PO_4 water solution (20 μL for plasma, 50 μL for BAL fluid) was added and vortexed for 45 s. Samples were mixed with 100 μL of 0.1% formic acid in acetonitrile/methanol (1:1) and vortexed. Then, samples (140 μL for plasma, 200 μL for BAL fluid) were transferred into a Phenomenex Protein Precipitation SPE plate. The filtrate was collected in a 1 mL 96-well plate and centrifuged at 2,000 rpm for 5 min. Samples were dried under nitrogen gas at 60°C, reconstituted with 150 μL of 0.1% formic acid in water/acetonitrile (95:5), placed into a cooled autosampler (10°C), and 5 μL injected into the UPLC-MS/MS.

Major polymyxin B components and the internal standard avibactam were resolved using a Waters UPLC Acquity system with a Waters Acquity CSH C_{18} column (100 mm × 2.1 mm; internal diameter,

1.7 μm, Milford; MA). The mobile phase contained 0.1% formic acid in water and acetonitrile in a gradient method. The flow rate was 0.500 mL/min, and the column temperature was 45°C. An API5000-Qtrap triple quadrupole mass spectrometer from Sciex (Foster City, CA, USA) equipped with a TurboIonSpray source was operated in the positive-ion mode. Optimal settings were as follows: ion-spray voltage, 5.5 kV; ion source temperature, 650°C; gas 1, 60 lb/in^2; gas 2, 60 lb/in^2; curtain gas, 30 lb/in^2; and CAD gas set as 6.00 lb/in^2. A multiple-reactions monitoring method was used for quantification using the Analyst 1.5.2 software. After separating the polymyxin B components by chromatography, the monitored mass/charge (*m/z*) transitions were *m/z* 402 → *m/z* 101 and *m/z* 402 → *m/z* 120 for polymyxin B1, *m/z* 402 → *m/z* 101 and *m/z* 402 → *m/z* 396 for polymyxin B1-Ile, *m/z* 398 → *m/z* 101 and *m/z* 398 → *m/z* 391 for polymyxin B2, *m/z* 398 → *m/z* 101 and *m/z* 398 → *m/z* 120 for polymyxin B3, and *m/z* 267 → *m/z* 186 for avibactam.

Urea concentrations in plasma ($Urea_{Plasma}$) and BAL fluid ($Urea_{BAL}$) were analyzed using UPLC-MS/MS (48). The ratio of these two urea concentrations defined the dilution factor between BAL fluid and ELF (9). The concentrations of individual polymyxin B components in ELF (C_{ELF}) were then calculated as $C_{BAL} \cdot Urea_{Plasma}/Urea_{BAL}$, where C_{BAL} is the concentration of the respective polymyxin B component in BAL fluid.

Population pharmacokinetic modeling. To propose informative doses and sampling times, a two-compartment PK model (i.e., a plasma and an ELF compartment) with first-order absorption and elimination was developed using an in-house data set (data not shown). The lung ELF compartment contained a saturable binding site for polymyxin B to model free and total drug.

We subsequently developed population PK models for the four polymyxin B components based on the dose range data (study 1) of the bloodstream and lung infection models. Absorption after subcutaneous dosing was described by first-order or Michaelis-Menten kinetics. Disposition was described by a linear one compartment model, plus a lung ELF compartment for the lung infection model. The volume of distribution of the ELF compartment was fixed to a small value (2 mL), to minimize its effect on the largely monoexponential plasma concentration profiles. Elimination was described by first-order, Michaelis-Menten, or parallel first-order plus Michaelis-Menten kinetics.

Our UPLC-MS/MS assay quantified the total drug concentration in ELF. While we did not observe unbound drug concentrations in ELF, we implemented a saturable binding model of polymyxin B in lung ELF. This allowed the model to mathematically distinguish between unbound and bound polymyxin B concentrations in ELF. This was accomplished by estimating the ratio (F_{ELF}) of th AUC for unbound polymyxin B concentrations in ELF and the total drug AUC in plasma. This ratio characterized the average extent of ELF penetration as implemented previously (49–51). Polymyxin B is known to bind bacterial LPS, which presents a potential "sink" (10, 22). Thus, a lung compartment with saturable binding of polymyxin B to bacterial LPS was evaluated. In lack of data on the amount of LPS in the mouse lung, we used a simplified model with a constant maximum binding capacity (i.e. rate constant, B_{max}), the polymyxin B amount (i.e. sum of the four components) in the lung yielding half of B_{max} (BC_{50}), and the half-life of dissociation (T_{off}).

For both the lung and the bloodstream infection model, we first developed a PK model for the sum of four polymyxin B components in plasma and ELF (Fig. S1) and then developed refined models for each of the four polymyxin B components based on data from studies 1 to 4. Humanized dosage regimens were subsequently developed and validated in study 2, followed by updating the population PK parameter estimates. Finally, PK data from efficacy studies 3 and 4 were included, and the final population PK model parameters were estimated by simultaneously fitting the data from all four studies (Fig. S1).

Models with the same or different clearances and volumes of distribution for the four components were considered. Moreover, we estimated models with the same or different clearances and volumes of distribution between the four studies of each infection model. This accounted for potential between-study differences. Finally, we estimated the parameters of the lung model with and without the ELF data to confirm potential differences in clearance between the lung and bloodstream infection models.

Models were evaluated based on the pharmacological plausibility of the model structure and parameter estimates as well as various diagnostic plots. All estimations were performed using the importance sampling algorithm (pmethod = 4) in the S-ADAPT (ver. 1.57) and SADAPT-TRAN software packages (52–54). Monte Carlo simulations were run in the Berkeley Madonna software (ver. 8.3.18; Univ. of California). A combined additive plus proportional model was used to describe the residual error in plasma and ELF.

Humanized dosage regimens. Clinical PK studies in critically ill patients were reviewed, and plasma concentration profiles were simulated via Monte Carlo simulations to obtain the median and 10th to 90th percentiles of polymyxin B concentrations in patients (40, 41). A loading dose of 2 mg/kg with a maintenance dose of 1.5 mg/kg every 12 h (starting at 12 h) (7) was chosen to develop humanized dosage regimens in mice (18, 39). Mean PK profiles for polymyxin B in infected mice were simulated based on the population mean PK parameters from the dose range studies (study 1) of the bloodstream and lung infection models.

We sought to match the peak and trough concentrations as well as the AUC of total polymyxin B in plasma between mice and humans. Doses, dosing intervals, and potential times of supplement doses in mice were adjusted for murine concentrations to stay within the 10th to 90th percentiles of simulated concentrations in patients. This approach accounted for potential differences in the number of exponential phases between mice and humans (e.g., monoexponential in mice versus biexponential in humans) and for the likely shorter disposition half-lives in mice. Humanized dosage regimens contained up to four polymyxin B doses per day in mice for feasibility reasons. Given the 12 h dosing interval in humans, the mouse dosage regimens comprised two main doses at 0 and 12 h, with or without two supplement doses at optimized intermediate time points (e.g., at 6 and 18 h). We are not aware of human PK data on the four polymyxin B components or of human ELF penetration data on polymyxin B. Thus, we matched

total drug concentrations in plasma and assumed that protein binding was comparable between mice and patients. Prior studies reported an unbound fraction of 8.6% in the plasma of mice infected by *Klebsiella pneumoniae* (19) and an unbound fraction of ~10% in patients (41, 42).

SUPPLEMENTAL MATERIAL

Supplemental material is available online only.

SUPPLEMENTAL FILE 1, PDF file, 1.9 MB.

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